

knowledge of the space sector and the underlying risks.

Various lunar rover missions that have so far taken place are given in the table below.

Without vision, no space programme can exist. China launched its rover Yutu 2 on December 7, 2018, from Xichang launch station. It successfully landed on Von Karman Crater. China's far-side moon rover is busy exploring its exotic new home. A part of Yutu 2 Chang'e 4 mission—in Chinese mythology, Chang'e 4 is a moon goddess and Yutu or Jade Rabbit is her pet—Yutu 2 is a six-wheeled rover, and is likely to cover 2500 kilometres of the south pole of moon Aitken Basin. Yutu 2 houses four scientific instruments that will be used to study the surrounding dirt and rocks, and to probe the far side's sub-surfaces. Such observations could help scientists better understand the moon's composition, structure and evolution.

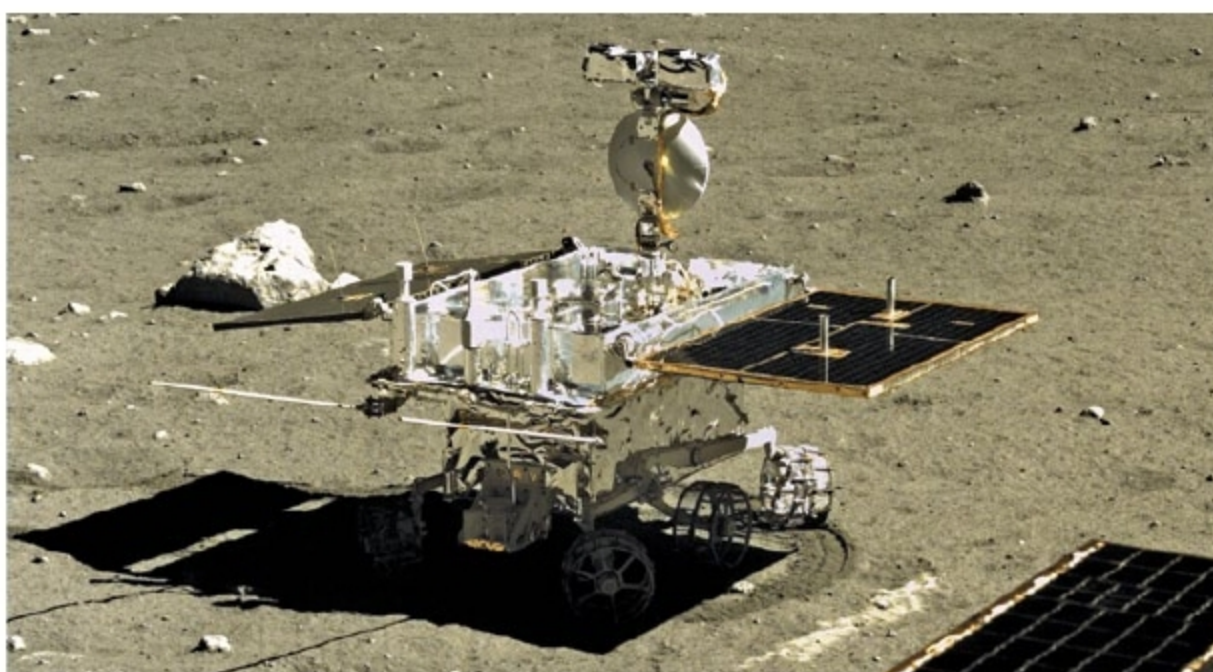
On moon, fourteen days are cold when temperature is around -173°C . Other fourteen days are hotter, and during this time temperature reaches as high as $+127^{\circ}\text{C}$. Yutu 2 has to face these extremities in temperature. Planting potato seeds and other seeds on the moon is one of the important experiments to be carried out by it.

Yutu 2 is carrying a radioscope, which is equipped with a thermoelectric generator in which radioactive substance is present, which is converted into electricity. Queqiuoa (Mas Pie Bridge) is a satellite rotating around the moon, and relaying data and commands from Yutu 2 to Earth.

In the rapidly transforming technological world, India is currently focusing its attention on emerging technologies. Though not by design, space rover technology is still an emerging sector in India.

Lunar Rover Missions

Name of rover	Country	Year
Lunokhod 1	Soviet Union	1969
Apollo Lunar	USA	1971
Lunokhod 2	Soviet Union	1973
Yutu 1	China	2013
Yutu 2	China	2018
Chandrayaan-2	India	2019



China's Yutu 2 lunar rover (Credit: www.universetoday.com)

Economic reforms like Make in India and startups from various sectors have done wonders in terms of productivity, consumer welfare and more. This includes the Indian space sector as well.

India's Chandrayaan-2 was scheduled for October 2018, but got postponed to 2019. The average-sized, indigenously-developed Indian rover started giving trouble with re-throttling in lander parts. This was due to problems with thrusters. Previously, GSLV II was proposed to ferry Chandrayaan-2.

Mars rover missions making reforms

Proper utilisation of rover technology will help provide better opportunities to explore Mars as well. Geographical exploration including study of mineral compositions, and environmental studies such as temperature and pressure (particularly wind) are some important studies to be made by rovers. Such findings may act as key instruments for landing on Mars.

Six rovers have been sent to Mars so far, of which four projects have been successful. Mars2 was sent in 1971, and its landing failed while

taking Prop-M with it. Mars3, sent in 1971, stopped communicating after about twenty seconds after landing, making the mission a failure.

Sojourner rover landed successfully on July 4, 1997. However, communications from it were stopped on September 27, 1997. Spirit Mars Exploration device-A (MER-A), launched on June 10, 2003, landed on Mars on January 4, 2004. It worked for nearly six years and travelled through 7.73 kilometres.

Opportunity Mars Exploration device-B (MER-B) was launched on July 7, 2003, and landed on Mars on January 25, 2004. It covered a total distance of 40.25 kilometres. On 2018, it stopped functioning when Mars dust-storms blocked the sunlight needed to recharge its batteries.

Curiosity rover was launched by NASA on November 26, 2011. It landed on Gale Crater on August 6, 2012. It is still operational.

Improved experiments by Curiosity. On November 26, 2011, NASA sent Curiosity rover to Mars. Examination of loose rocks, sand and dust by Curiosity has provided great understandings about local and global soil details of the red planet. A key finding of Curiosity is that, water molecules on Mars are bound to fine-grained soil particles at Gale Crater, where it landed. Findings about both crystalline and non-crystalline components in soil provide information about the planet's volcanic history.

Curiosity is 3.05m (10-feet) long, 2.74m (9-feet) wide and 2.13m (7-feet) tall, and its arm reaches 2.13m

Planned Rover Missions to Mars

- Rosalind Franklin by combined efforts from Europe and Russia
- Mars 2020 (Maven) by NASA
- Mars global remote sensing orbiter and small rover by China